Experiment 3

Survey Report on Emerging Trends in Low-Code Application Development

Aim

To study and analyse the emerging trends in low-code application development and understand how these technologies are used in developing the Transport Management System using Zoho Creator.

Theory

Low-code application development is continuously evolving with new technologies and features that simplify application development. Zoho Creator provides several capabilities that support modern business application development. The major emerging trends relevant to the Transport Management System are:

- Artificial Intelligence (AI) Integration: AI can help analyse transport data, predict maintenance requirements, and improve decision-making.
- Citizen Development: Non-technical users can develop and manage applications using drag-and-drop tools with minimal coding.
- Cloud-Based Development: Zoho Creator provides cloud-based application development, allowing administrators to access transport records from different locations.
- Business Process Automation: Workflows automate vehicle status updates, trip completion processes, and insurance expiry notifications.
- Cross-Platform Development: Applications can be accessed through web browsers and supported mobile devices.
- API and Third-Party Integration: Zoho Creator supports integration with external services for notifications, data sharing, and other transport-related operations.
- Enhanced Security and Access Control: User permissions and role-based access help protect vehicle, driver, passenger, and trip information.
- Data Visualization and Analytics: Dashboards provide KPI cards, pie charts, donut charts, and bar charts to monitor transport operations.

Procedure

1. Study the concept of low-code application development and its importance in modern software development.
2. Identify the features of Zoho Creator used for developing the Transport Management System.
3. Collect information about emerging trends from reliable sources such as official documentation and technical resources.
4. Analyse trends such as AI integration, cloud-based development, workflow automation, cross-platform access, and data visualization.
5. Study how workflow automation helps update vehicle status during trip starting and completion.
6. Examine how dashboards and reports help monitor vehicles, drivers, passengers, and trips.
7. Compare the benefits and challenges of these technologies in transport management.
8. Record the observations and prepare the survey report based on the findings.

Result

The survey revealed that low-code application development is rapidly evolving with trends such as AI-assisted development, cloud-native platforms, citizen development, business process automation, API integration, and enhanced security. These advancements are improving development speed, reducing costs, and enabling organizations to build scalable applications more efficiently while supporting digital transformation initiatives.